

Approved Gene & Cell Therapies — Memorisation Sheet

Therapy	What to memorise it as	Disease / use	Vector / platform	Main tissue target	Why it matters
Luxturna	Retina AAV gene replacement	RPE65 inherited retinal dystrophy	AAV2	Retina / RPE	Classic local AAV gene replacement precedent
Zolgensma	Paediatric neuromuscular AAV gene replacement	Spinal muscular atrophy	AAV9-SMN1	Motor neurons / CNS / peripheral motor system	Best-known systemic AAV9 childhood disease precedent
Libmeldy / Lenmeldy	Lentiviral HSC therapy for CNS leukodystrophy	Metachromatic leukodystrophy	Ex vivo lentiviral HSC, ARSA	Haematopoietic + CNS benefit	Key precedent for CNS disease treated via modified stem cells
Skysona	Lentiviral HSC therapy for cerebral ALD	Cerebral adrenoleukodystrophy	Ex vivo lentiviral HSC, ABCD1	Haematopoietic + CNS	Important leukodystrophy precedent
Strimvelis	Ex vivo immune gene therapy	ADA-SCID	Ex vivo retroviral / lentiviral-style HSC correction	Immune / haematopoietic	Classic inherited immune deficiency precedent
Zynteglo	Lentiviral HSC therapy for beta-thalassaemia	Transfusion-dependent beta-thalassaemia	Ex vivo lentiviral HSC, HBB	Blood / haematopoietic	Key haemoglobinopathy gene-addition precedent
Casgevy	CRISPR-edited HSC therapy	Sickle cell disease / beta-thalassaemia	Ex vivo CRISPR editing of HSCs	Blood / haematopoietic	Landmark approved CRISPR gene-editing precedent
Hemgenix	AAV liver gene therapy for secreted clotting factor	Haemophilia B	AAV5-FIX / F9	Liver → blood	Classic liver-as-factory precedent
Roctavian	AAV liver gene therapy for haemophilia A	Haemophilia A	AAV5-FVIII / F8	Liver → blood	Similar to Hemgenix but harder because F8 is larger
Elevidys	AAV muscle microgene therapy	Duchenne muscular dystrophy	AAVrh74 micro-dystrophin	Skeletal / cardiac muscle	Key large-gene / muscle precedent
Upstaza	Direct brain AAV therapy	AADC deficiency	AAV2-DDC	Brain / CNS	Important direct CNS delivery precedent
Vyjuvek	Topical repeat-dose HSV gene therapy	Dystrophic epidermolysis bullosa	HSV1-COL7A1	Skin	Shows local, repeatable gene delivery
Kymriah	First landmark CAR-T	B-cell cancers	Ex vivo CD19 CAR-T	Immune / cancer	Know as cancer cell-therapy archetype
Yescarta	Major CAR-T lymphoma therapy	B-cell lymphoma	Ex vivo CD19 CAR-T	Immune / cancer	Another major oncology GTMP precedent
Tecartus	CAR-T for mantle cell / ALL	B-cell cancers	Ex vivo CD19 CAR-T	Immune / cancer	Useful oncology precedent
Abecma	BCMA CAR-T	Multiple myeloma	Ex vivo BCMA CAR-T	Plasma cells / cancer	Myeloma CAR-T archetype
Carvykti	BCMA CAR-T	Multiple myeloma	Ex vivo BCMA CAR-T	Plasma cells / cancer	High-impact myeloma CAR-T precedent
Breyanzi	CD19 CAR-T	B-cell lymphoma / leukaemia	Ex vivo CD19 CAR-T	Immune / cancer	Another approved CAR-T example
Aucatzyl	CD19 CAR-T	Acute lymphoblastic leukaemia	Ex vivo CD19 CAR-T	Immune / cancer	Recent CAR-T precedent
Glybera	Historic first EU AAV gene therapy	Lipoprotein lipase deficiency	AAV1-LPL	Muscle / lipid metabolism	Historically important, but withdrawn — not current

Colour key: ■ AAV (retina/CNS/brain) ■ HSC lentiviral ■ CRISPR-HSC ■ AAV liver (haemophilia) ■ AAV muscle ■ HSV skin ■ CAR-T ■ Historic / withdrawn